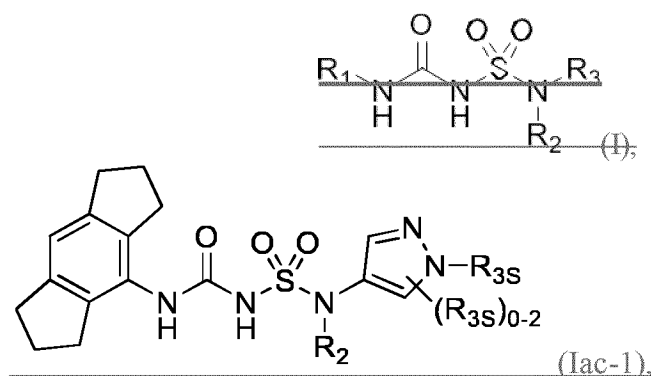


AUXILIARY REQUEST 2

CLAIMS:

1. A compound of Formula (Iac1):



or a solvate, or pharmaceutically acceptable salt thereof, wherein:

~~R₁ is C₃-C₁₆ cycloalkyl optionally substituted by one or more R_{1s}, wherein the C₃-C₁₆ cycloalkyl is a C₃-C₈ monocyclic cycloalkyl or polycyclic cycloalkyl, or, R₁ is a phenyl substituted by two or three R_{1s}, wherein each R_{1s} is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₁-C₆ alkoxy, C₁-C₆ haloalkoxy, or halo, and at least one R_{1s} is C₁-C₆ alkyl, C₁-C₆ haloalkyl, or halo;~~

~~R₂ is -(CHX₂X₂)_n-R_{2s}, wherein n is 0, 1, or 2, and each X₂ is independently H, C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl, wherein the C₁-C₆ alkyl, C₂-C₆ alkenyl, or C₂-C₆ alkynyl is optionally substituted with one or more halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo;~~

~~R_{2s} is unsubstituted 5 or 6- to 8-membered monocyclic heterocycloalkyl optionally substituted with one or more C₁-C₆ alkyl, C₂-C₆ alkenyl, C₂-C₆ alkynyl, C₁-C₆ haloalkyl, halo, -CN, -OH, -O(C₁-C₆ alkyl), -NH₂, -NH(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, or oxo; and~~

~~R₃ is 7- to 12-membered heterocycloalkyl or 5- or 6-membered heteroaryl optionally substituted with one or more R_{3s}, wherein each R_{3s} is independently C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with -O(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, halo, or -CN.~~

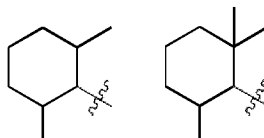
2. The compound of claim 1, wherein ~~R₁ is C₃-C₇ monocyclic cycloalkyl, C₉-C₁₀ bicyclic~~

cycloalkyl, C₁₂-C₁₆ tricyclic cycloalkyl, or phenyl, wherein the C₃-C₇ monocyclic cycloalkyl, C₉-C₁₀ bicyclic cycloalkyl, or C₁₂-C₁₆ tricyclic cycloalkyl is optionally substituted by one or more R_{1s}, the phenyl is substituted by two or three R_{1s}, and wherein at least one R_{1s} is C₁-C₆ alkyl, C₁-C₆ haloalkyl, or halo;

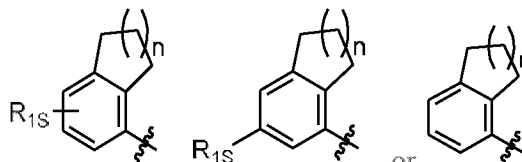
optionally wherein:

(a) the C₃-C₇ monocyclic cycloalkyl is cyclopentyl, cyclohexyl, or cycloheptyl, wherein the cyclopentyl, cyclohexyl, or cycloheptyl is optionally substituted by one or more R_{1s}; further

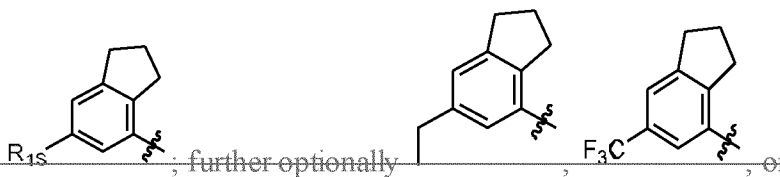
optionally wherein the C₃-C₇ monocyclic cycloalkyl is



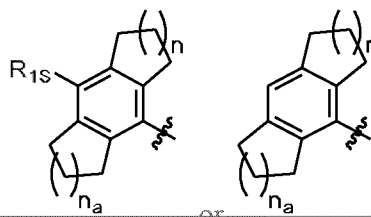
(b) the C₉-C₁₀ bicyclic cycloalkyl is



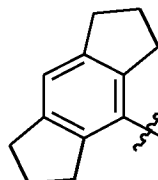
is 0, 1, or 2; optionally

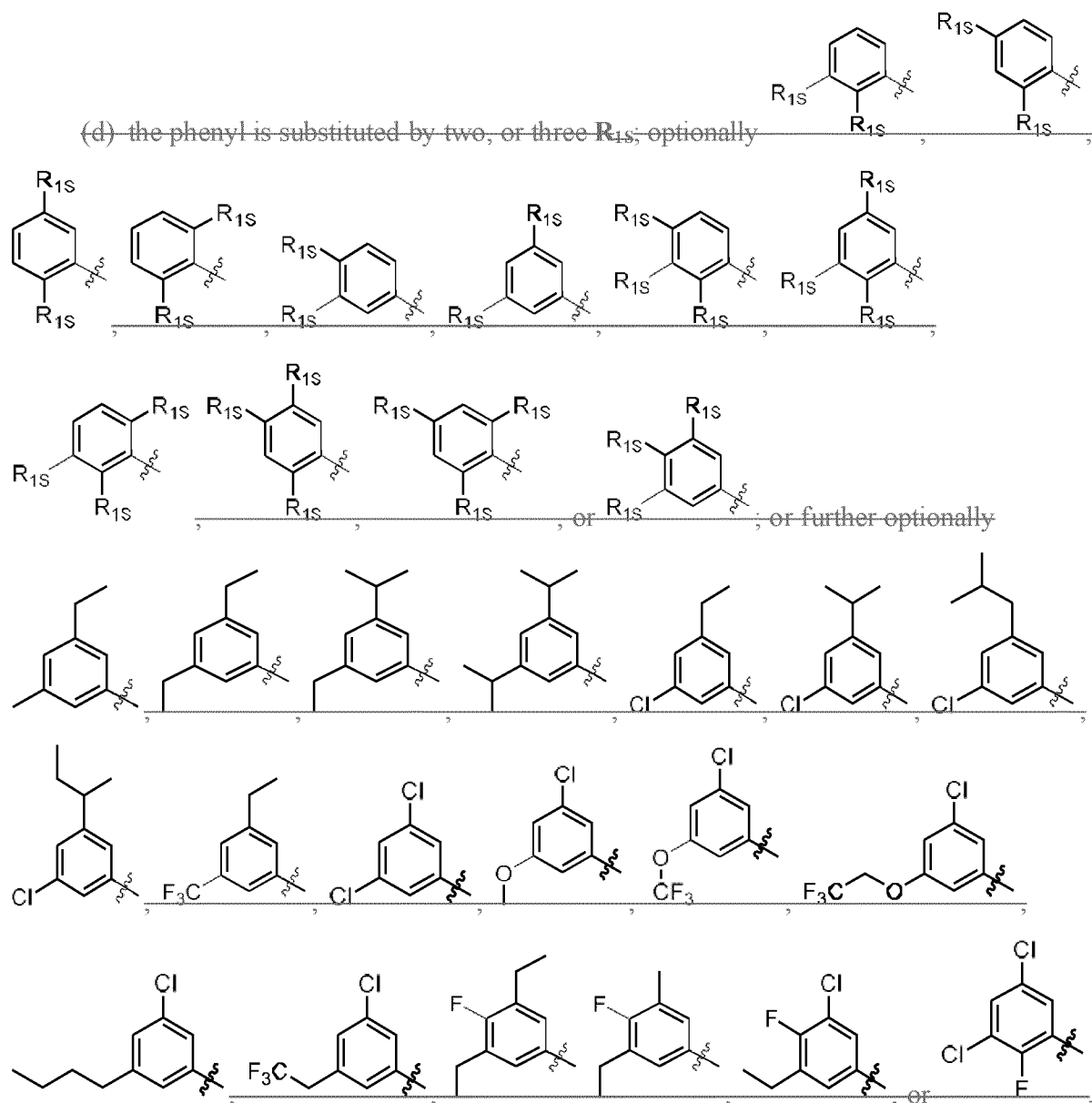


(c) the C₁₂-C₁₆ tricyclic cycloalkyl is



each independently are 0, 1, 2, or 3; further optionally

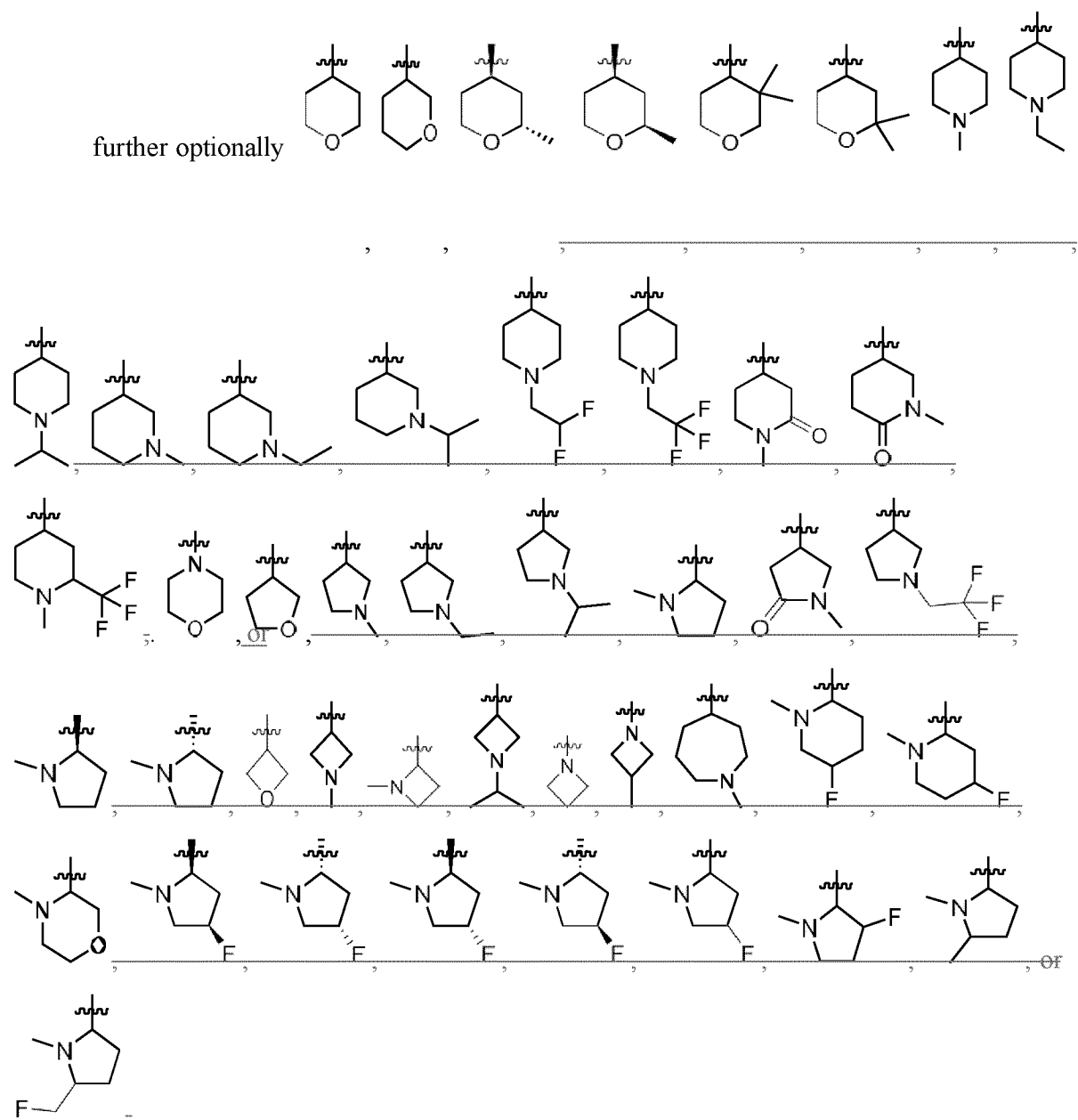




3. ~~The compound of claim 1 or 2, wherein R_2 is R_{2s} , (CX_2X_2) R_{2s} , or $(CX_2X_2)_2$ R_{2s}~~

42. ~~The compound of any one of claims 1 to 3, wherein R_{2s} is 4- to 8-membered monocyclic heterocycloalkyl optionally substituted with one or more C_1 - C_6 alkyl, C_2 - C_6 alkenyl, C_2 - C_6 alkynyl, C_1 - C_6 haloalkyl, halo, CN, OH, $O(C_1$ - C_6 alkyl), NH_2 , $NH(C_1$ - C_6 alkyl), $N(C_1$ - C_6 alkyl) $_2$, or oxo;~~

Optionally oxetanyl, azetidiny, tetrahydrofuranyl, pyrrolidinyl, tetrahydropyranyl, piperidinyl, or morpholinyl, or azepanyl, wherein the oxetanyl, azetidiny, tetrahydrofuranyl, pyrrolidinyl, tetrahydropyranyl, piperidinyl, morpholinyl, or azepanyl is optionally substituted with one or more C₁-C₆ alkyl, C₁-C₆ haloalkyl, or oxo; or

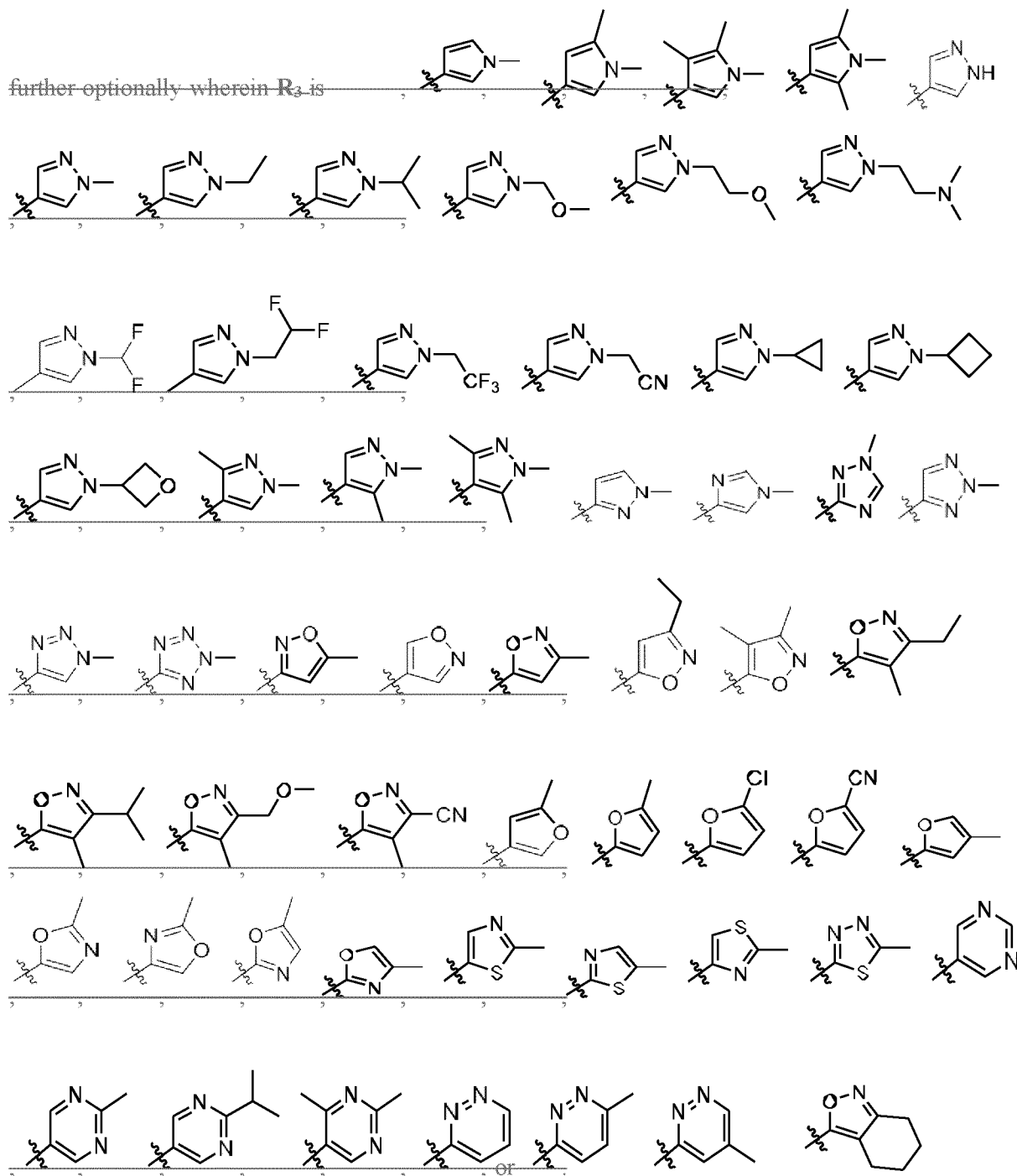


5. The compound of any one of claims 1 to 4, wherein **R₃** is 7 to 12 membered heterocycloalkyl optionally substituted with one or more **R_{3s}**; optionally wherein **R₃** is pyrrolyl, pyrazolyl, imidazolyl, triazolyl, tetrazolyl, isoxazolyl, furanyl,

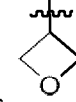
| ~~oxazolyl, isothiazolyl, thiadiazolyl, pyridinyl, pyrimidinyl, pyridazinyl, or 4,5,6,7-~~

tetrahydrobenzo[e]isoxazole, wherein the pyrrolyl, pyrazolyl, imidazolyl, triazolyl, tetrazolyl, isoxazolyl, furanyl, oxazolyl, isothiazolyl, thiadiazolyl, pyridinyl, pyrimidinyl, pyridazinyl, or 4,5,6,7 tetrahydrobenzo[e]isoxazole is optionally substituted with one or more R_3 s; or

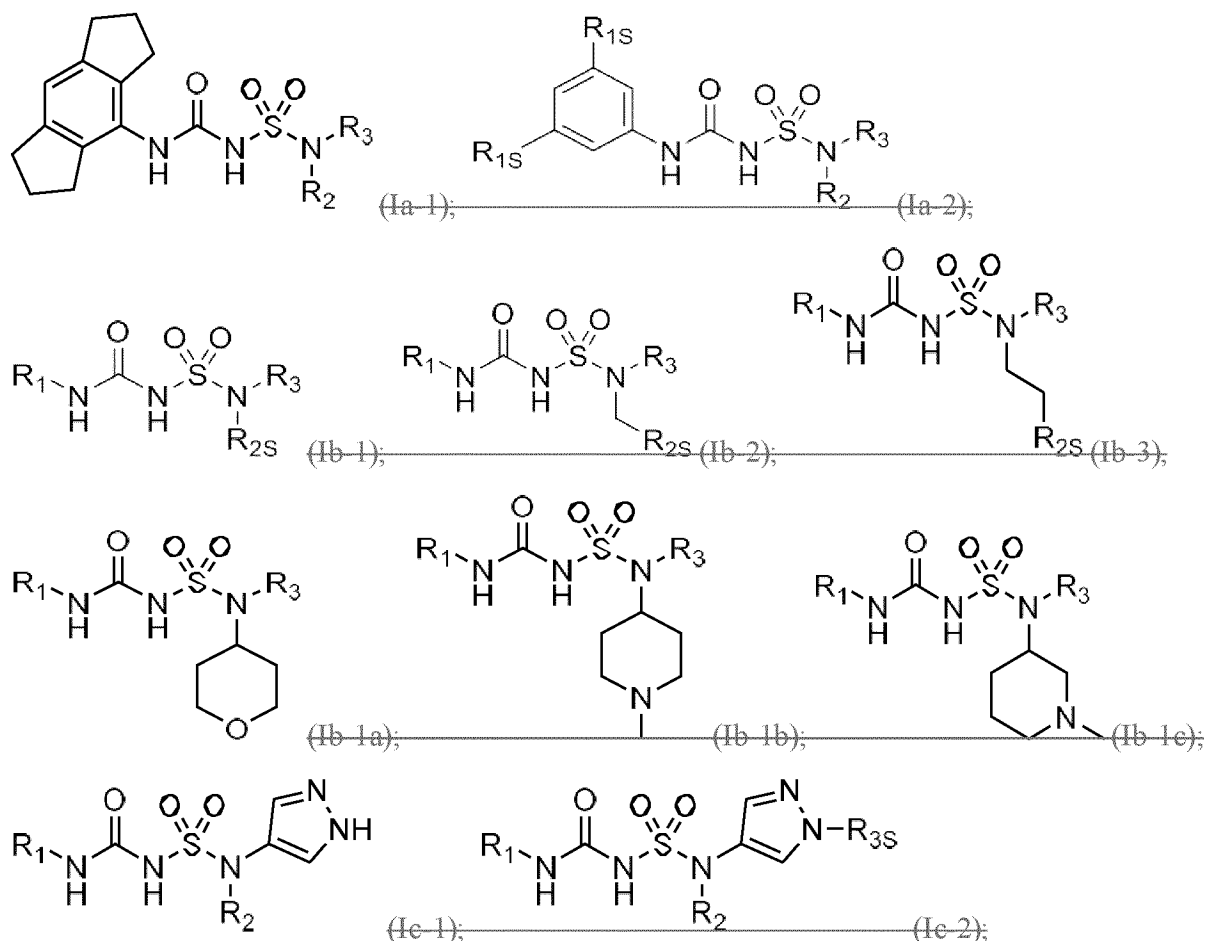
further optionally wherein R_3 is

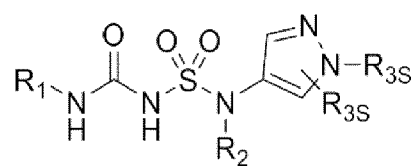


36. The compound of ~~any one of claims 1 or claim 2 to 5~~, wherein at least one R_{3s} is C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, halo, or C₃-C₈ heterocycloalkyl wherein the C₁-C₆ alkyl, C₁-C₆ haloalkyl, C₃-C₈ cycloalkyl, or C₃-C₈ heterocycloalkyl is optionally substituted with -O(C₁-C₆ alkyl), -N(C₁-C₆ alkyl)₂, halo, or -CN;

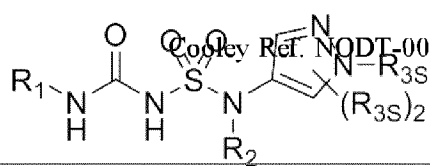
optionally wherein at least one R_{3s} is , , , methyl, ethyl, isopropyl, -CH₂OCH₃, -CH₂CF₃, -CH₂CH₂OCH₃, -CH₂CN, CH₂CF₂, or -CH₂CH₂N(CH₃)₂.

74. The compound of any one of the preceding claims, wherein the compound is of Formula ~~(Ia 1), (Ia 2), (Ib 1), (Ib 2), (Ib 3), (Ib 1a), (Ib 1b), (Ib 1c), (Ic 1), (Ic 2), (Ic 3), (Ic 4), (Ic 5), (Ic 6), (Ic 7), (Ic 8), (Iab 1), (Iab 2), (Iac 1), (Iac 2), (Iac 3), (Iac 4), (Ibc 1), (Ibc 2), (Iabc-1), or (Iabc-1a), or (Iabc-1b)~~:



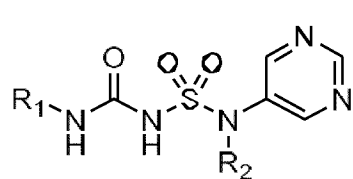


(Ic-3);

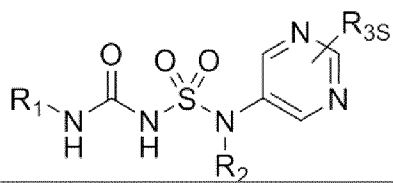


(Ic-4);

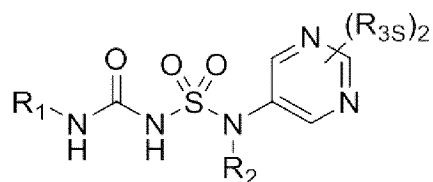
Cooley Ref. NODT-004/N01EP (330150-2065)



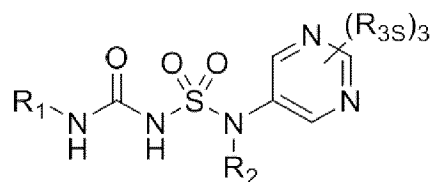
(Ie-5);



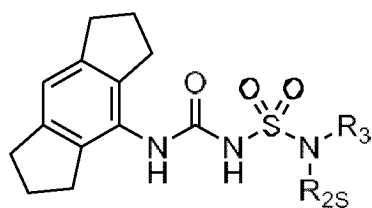
(Ie-6);



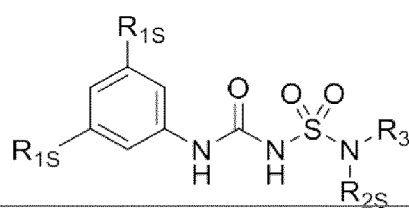
(Ie-7);



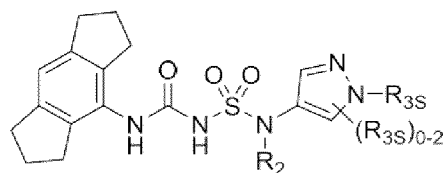
(Ie-8);



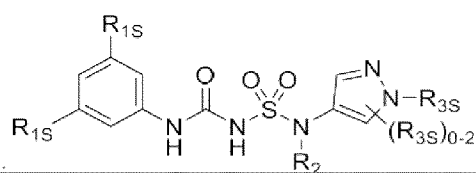
(Iab-1);



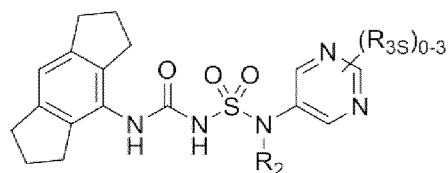
(Iab-2);



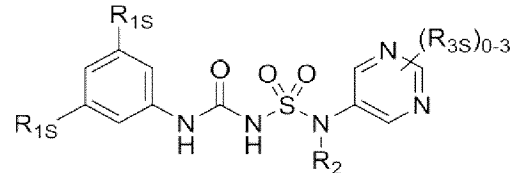
(Iac-1);



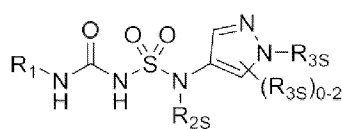
(Iac-2);



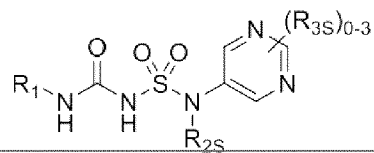
(Iac-3);



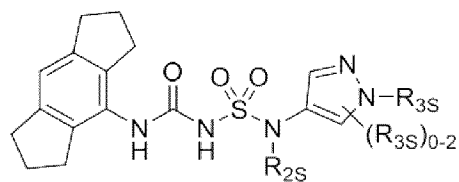
(Iac-4);



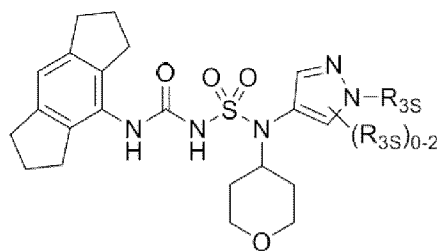
(Ibc-1);



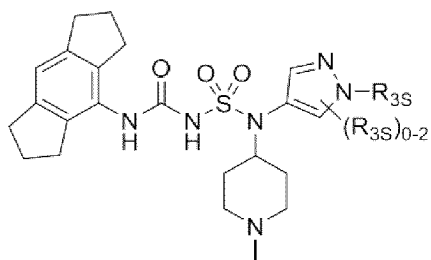
(Ibc-2);



(Iabc-1);



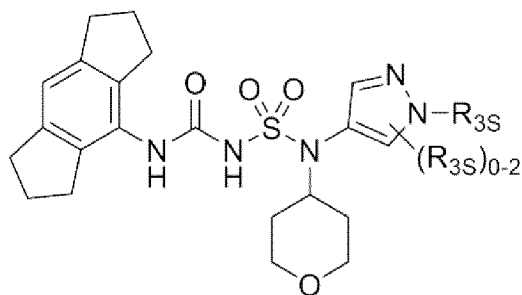
(Iabc-1a);



(Iabc-1b)

or a solvate, or pharmaceutically acceptable salt thereof, wherein ~~R₁~~, ~~R_{2S}~~, ~~R₂~~, and ~~R_{3S}~~ are as described ~~herein~~ in claim 1.

85. The compound of any one of claims 1 to ~~74~~, wherein the compound is of Formula (Iabc-1a):



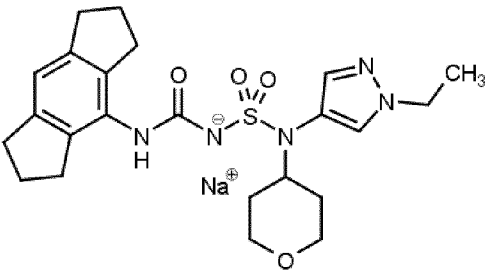
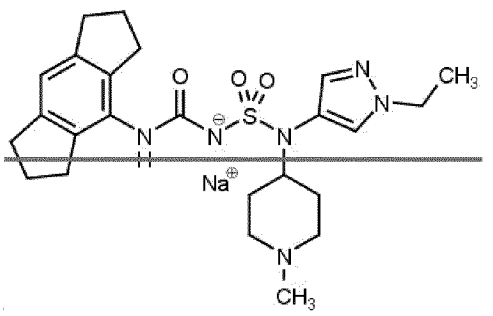
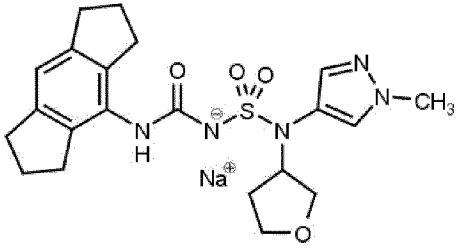
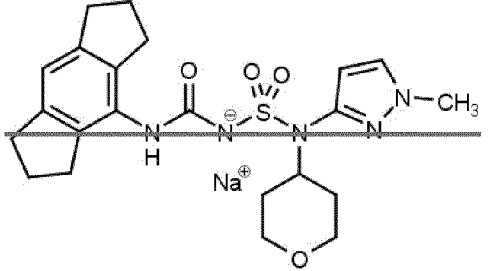
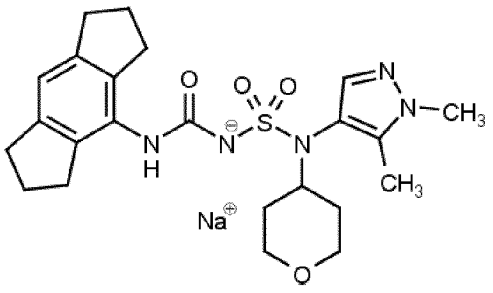
(Iabc-1a)

or a solvate, or pharmaceutically acceptable salt thereof, wherein **R_{3S}** is as described ~~herein~~ in claim 1.

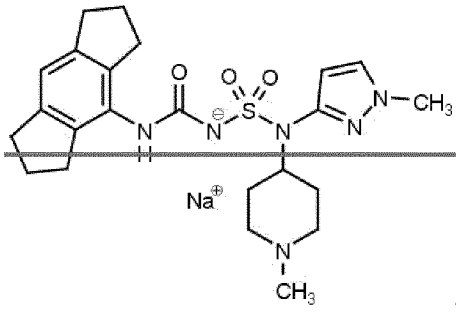
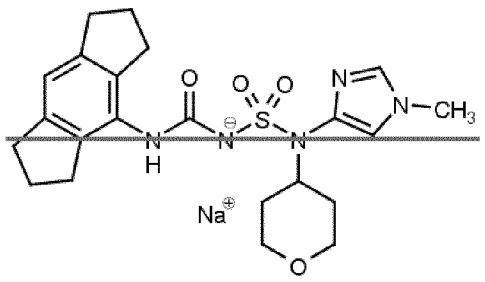
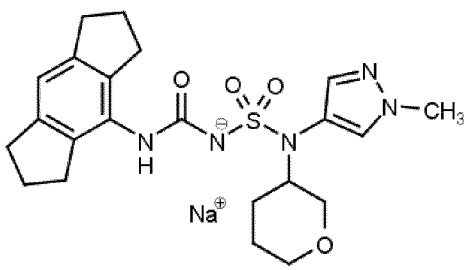
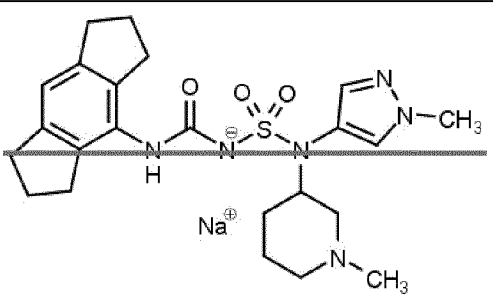
96. The compound of any one of ~~the~~ claims 1 to 85, being selected from:

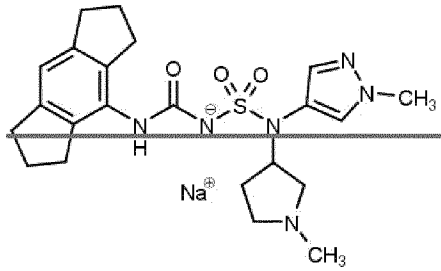
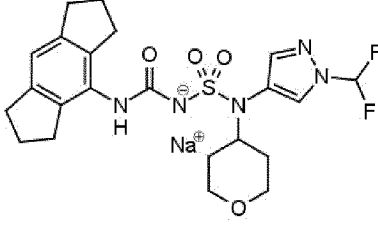
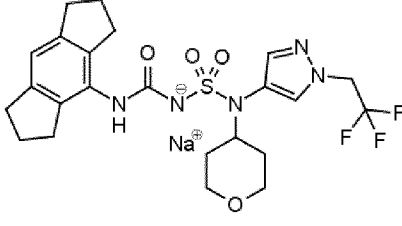
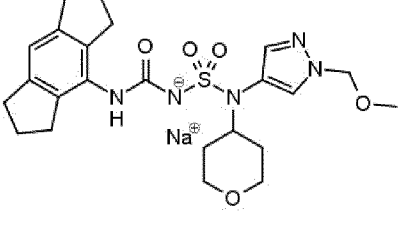
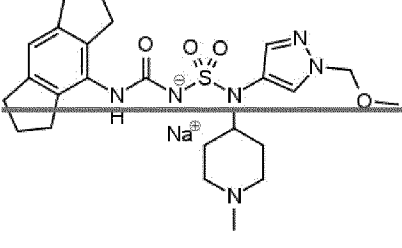
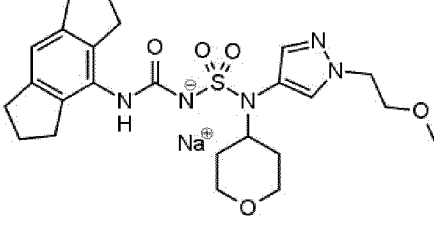
Compound No.	Structure
1	

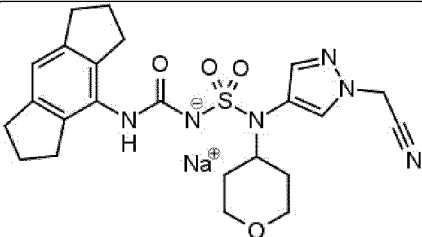
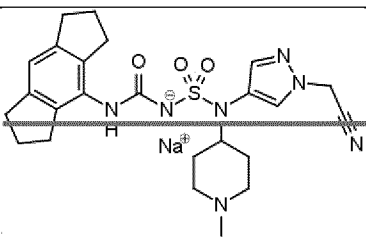
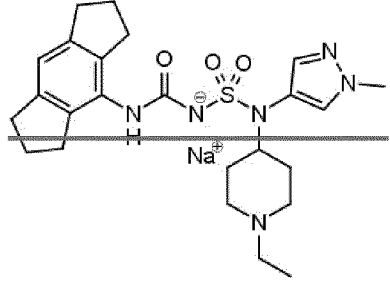
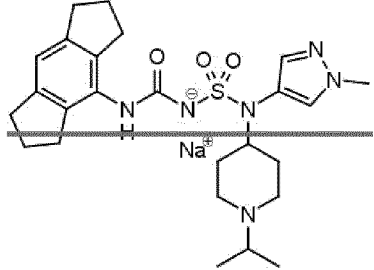
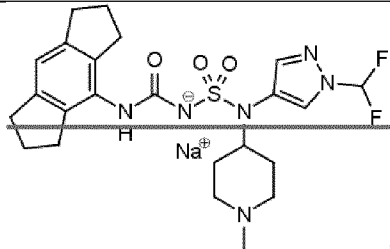
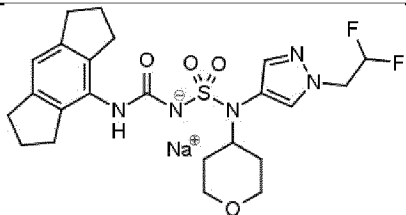
Compound No.	Structure
2	
3	
4	
5	

Compound No.	Structure
6	
7	
8	
9	
10	

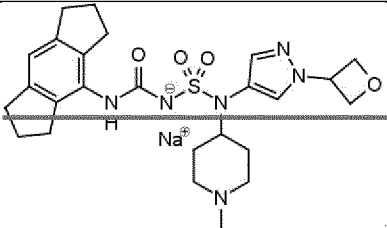
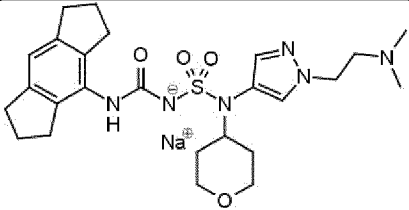
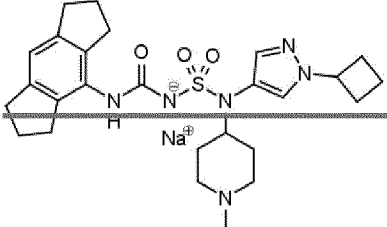
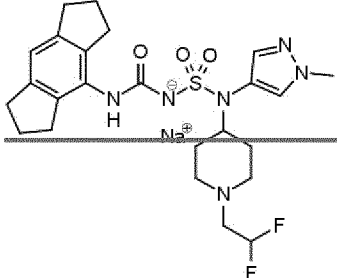
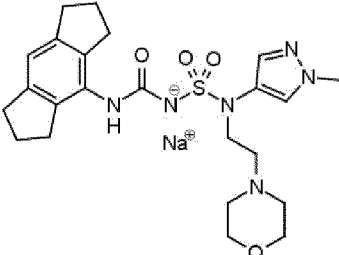
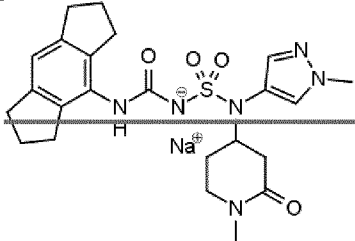
Compound No.	Structure
15	 <chem>CN1C=CN(C1)N(S(=O)(=O)N2C(=O)N3C4C5C6C7C8C9C5C4C3C2C6C7C8C9)C10CCN(C)CC10.[Na+]</chem>
16	 <chem>CN1C=CN(C1)N(S(=O)(=O)N2C(=O)N3C4C5C6C7C8C9C5C4C3C2C6C7C8C9)C10C11OC11C10.[Na+]</chem>
17	 <chem>CN1C=CN(C1)N(S(=O)(=O)N2C(=O)N3C4C5C6C7C8C9C5C4C3C2C6C7C8C9)C10C11OC11C10.[Na+]</chem>
18	 <chem>CN1C=CN(C1)N(S(=O)(=O)N2C(=O)N3C4C5C6C7C8C9C5C4C3C2C6C7C8C9)C10C11OC11C10.[Na+]</chem>
19	 <chem>CN1C=CN(C1)N(S(=O)(=O)N2C(=O)N3C4C5C6C7C8C9C5C4C3C2C6C7C8C9)C10C11OC11C10.[Na+]</chem>

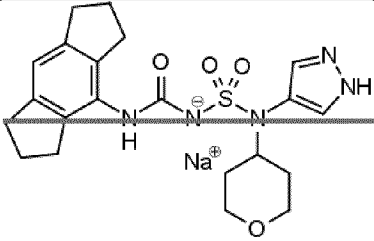
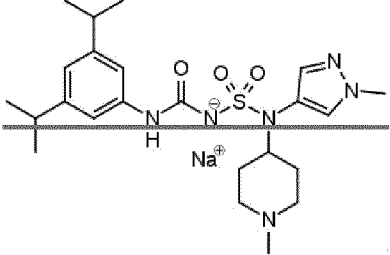
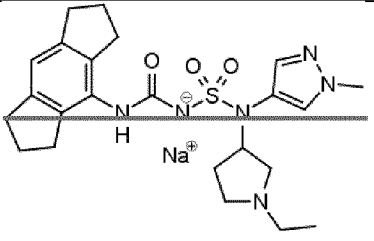
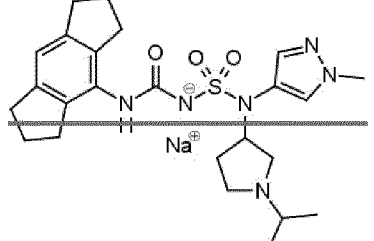
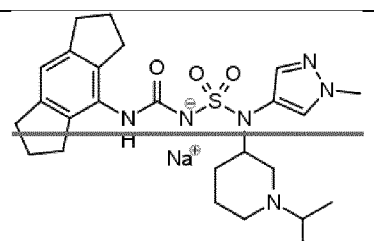
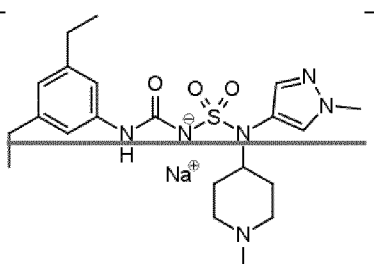
Compound No.	Structure
20	
21	
22	
23	

Compound No.	Structure
24	
25	
26	
27	
28	
29	

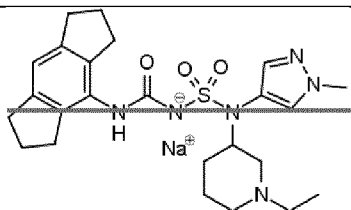
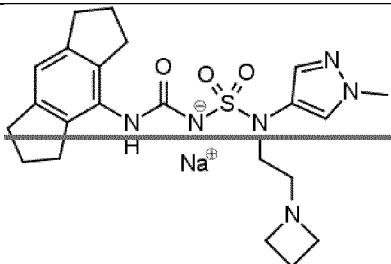
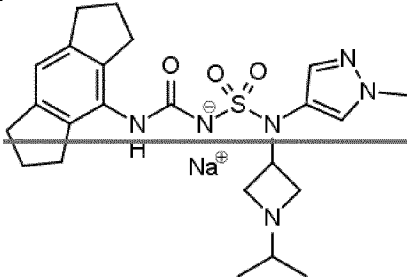
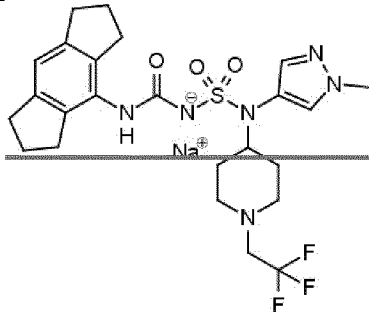
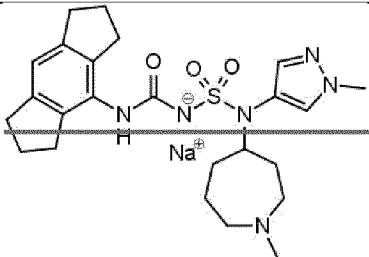
Compound No.	Structure
30	
31	
32	
33	
34	
35	

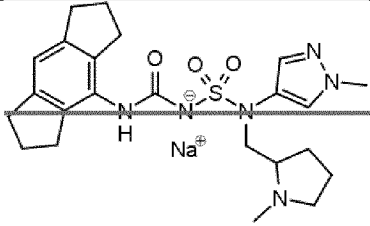
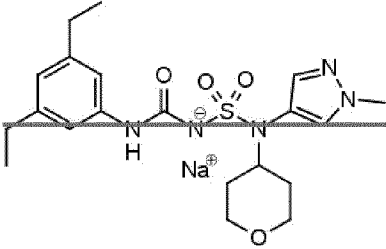
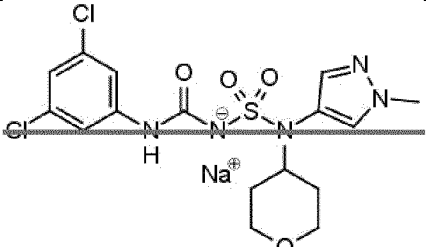
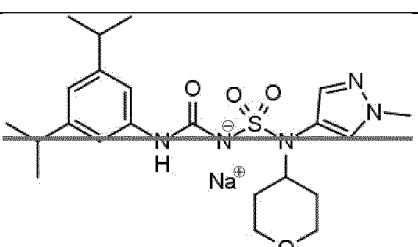
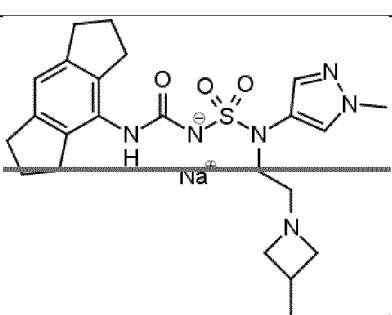
Compound No.	Structure
36	
37	
38	
39	
40	
41	

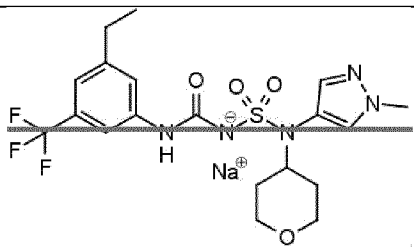
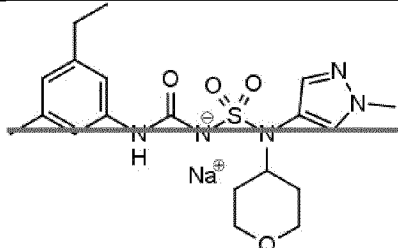
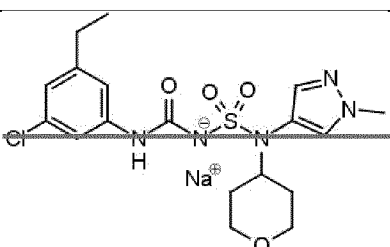
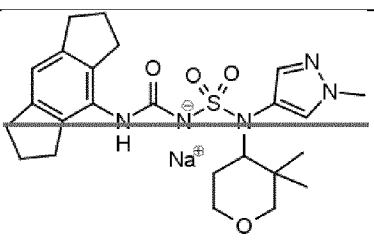
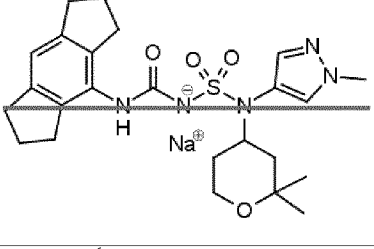
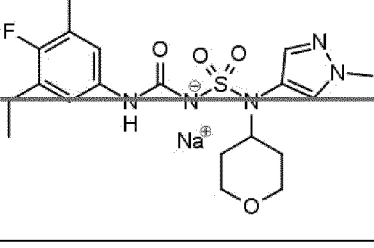
Compound No.	Structure
42	
43	
44	
45	
46	
47	

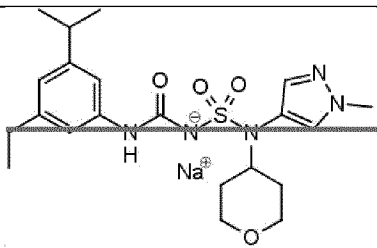
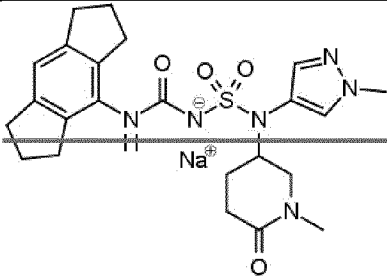
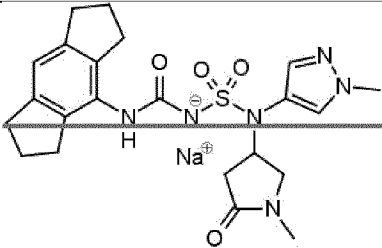
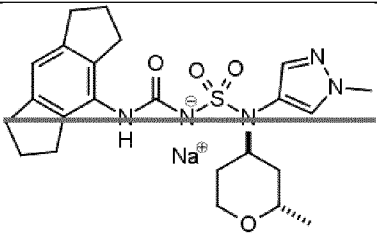
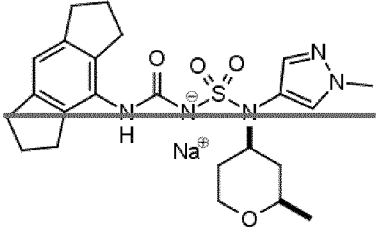
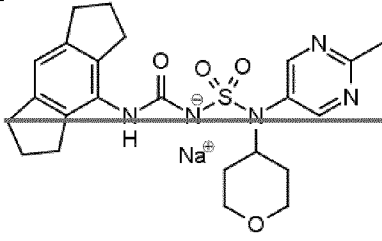
Compound No.	Structure
48	
49	
50	
51	
52	
53	

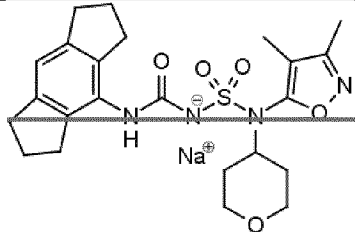
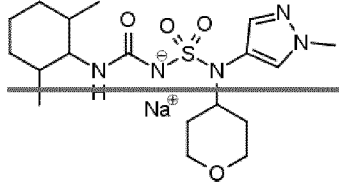
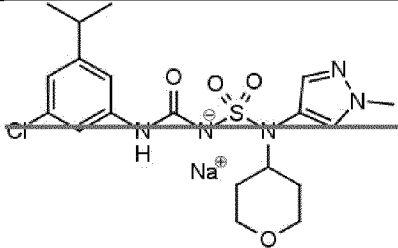
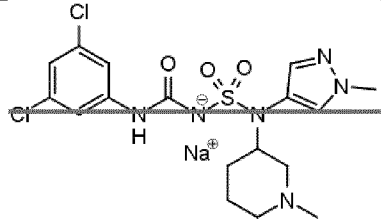
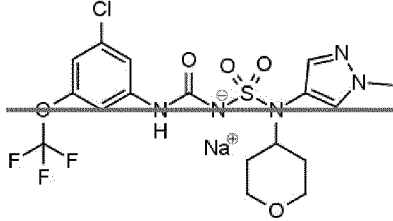
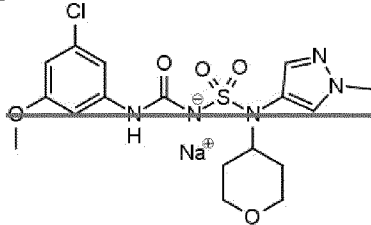
Compound No.	Structure
55	
56	
57	
59	
60	
61	

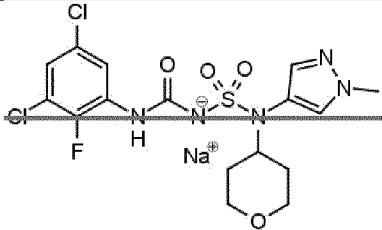
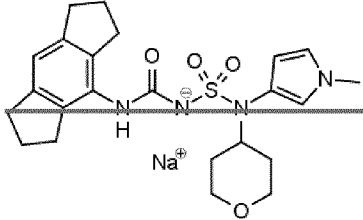
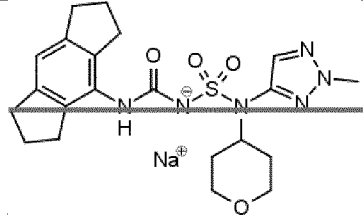
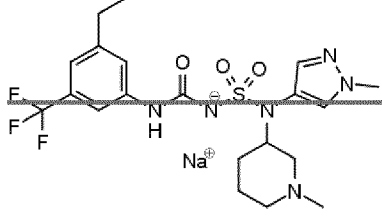
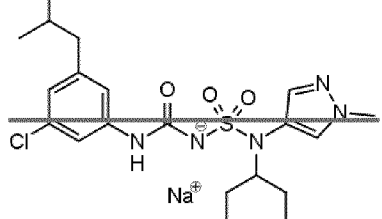
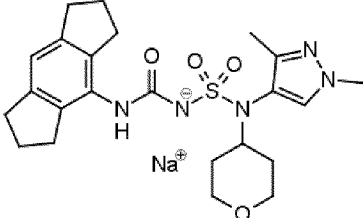
Compound No.	Structure
62	
63	
64	
65	
67	

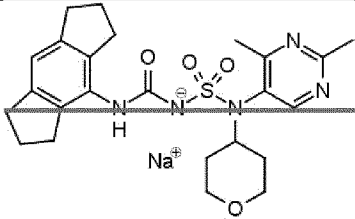
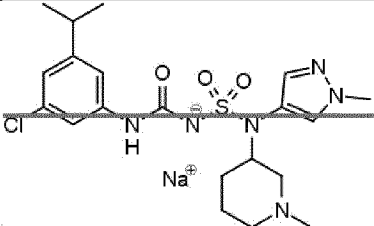
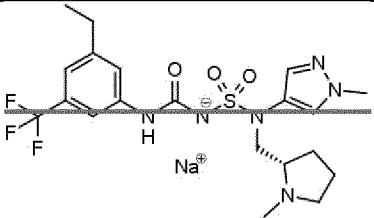
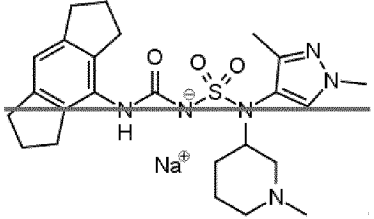
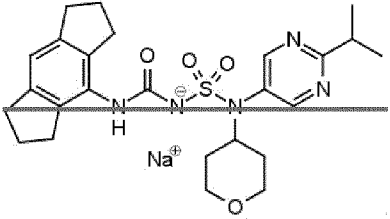
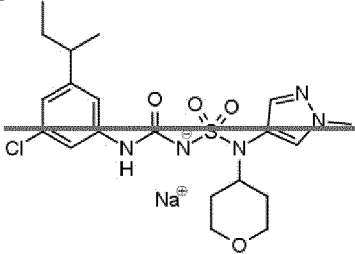
Compound No.	Structure
68	
70	
72	
73	
74	

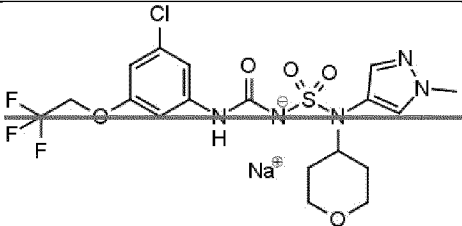
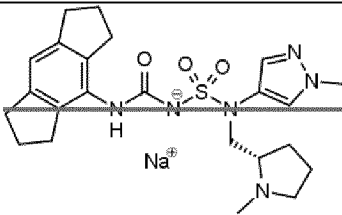
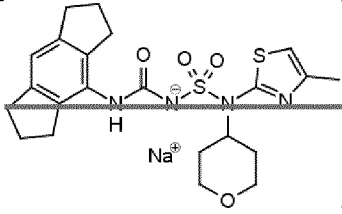
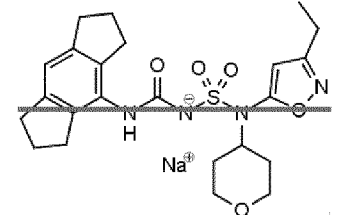
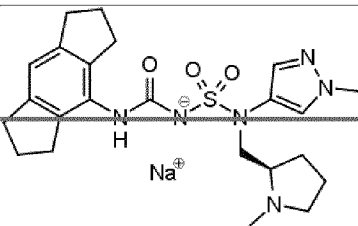
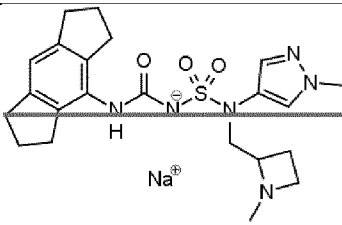
Compound No.	Structure
75	
76	
77	
78	
79	
80	

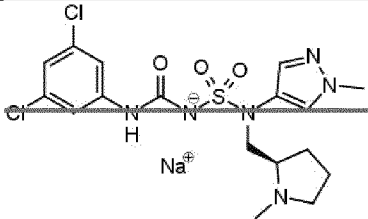
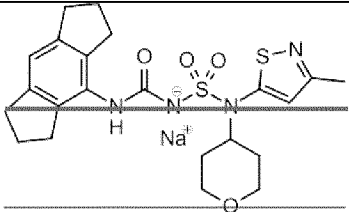
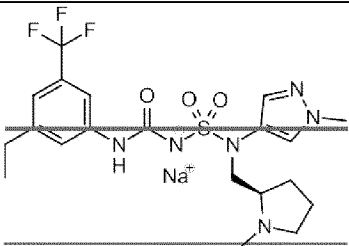
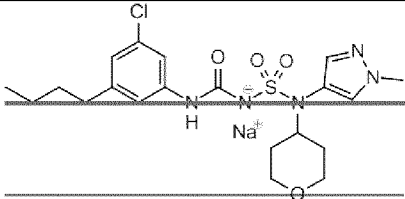
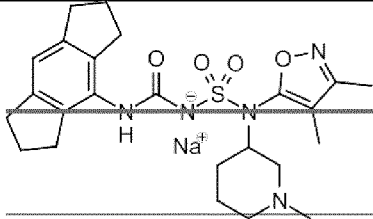
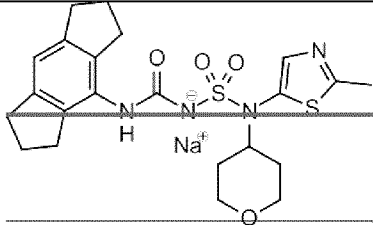
Compound No.	Structure
81	
82	
83	
84	
85	
86	

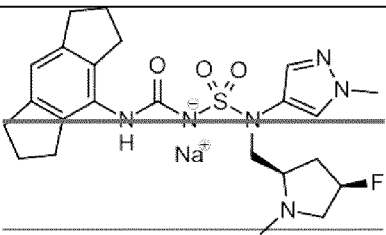
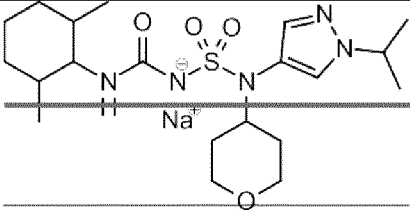
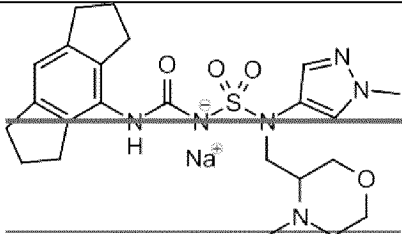
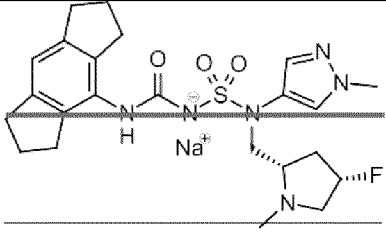
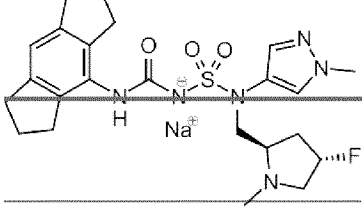
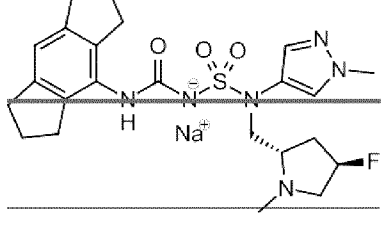
Compound No.	Structure
87	
88	
89	
91	
92	
93	

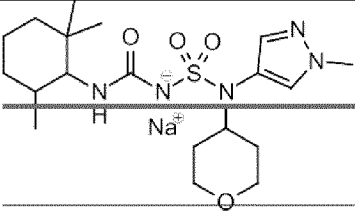
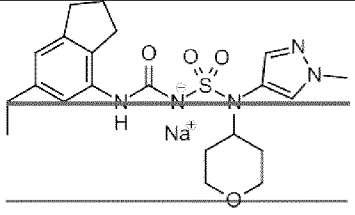
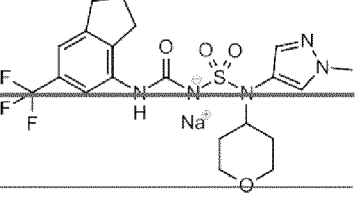
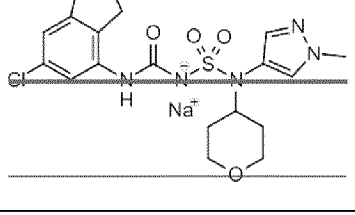
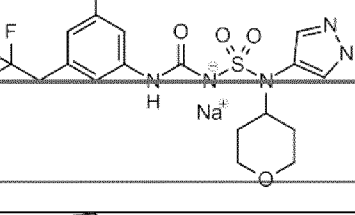
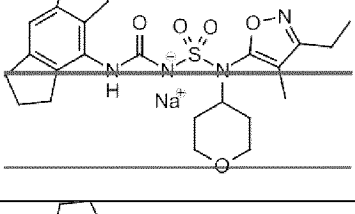
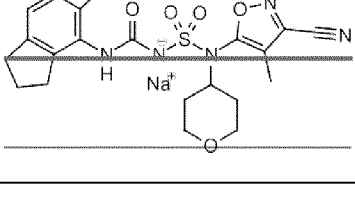
Compound No.	Structure
94	
95	
96	
97	
98	
99	

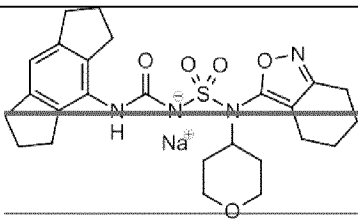
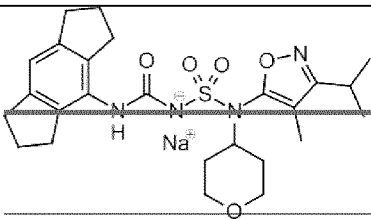
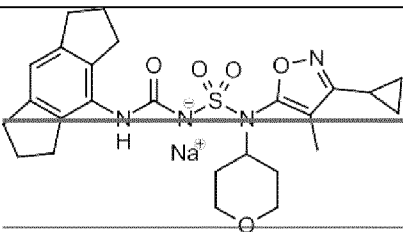
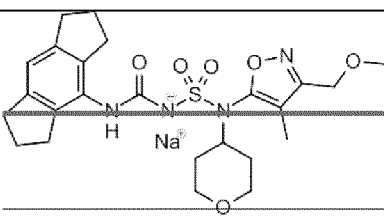
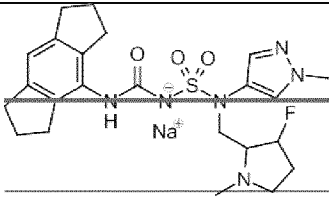
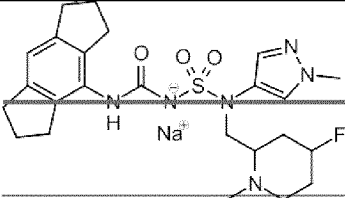
Compound No.	Structure
100	
101	
102	
103	
104	
105	

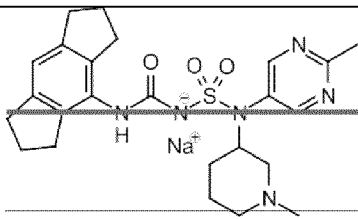
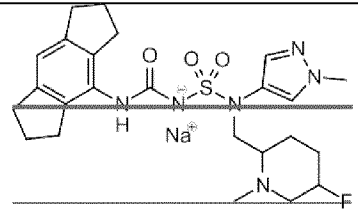
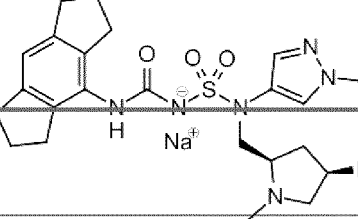
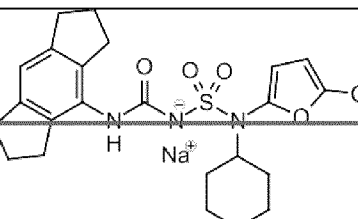
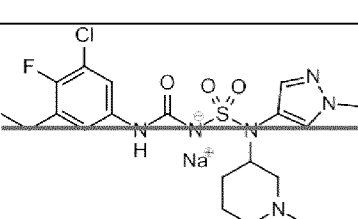
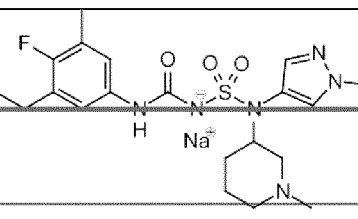
Compound No.	Structure
106	
107	
108	
109	
110	
111	

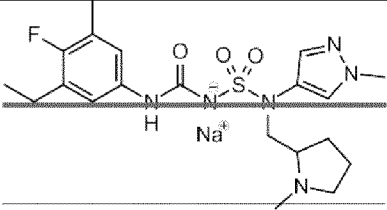
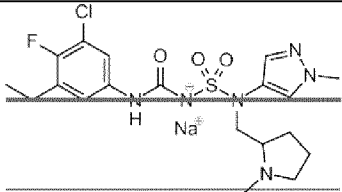
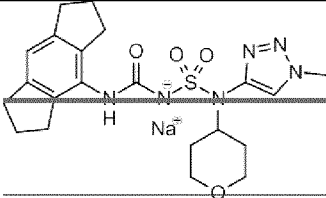
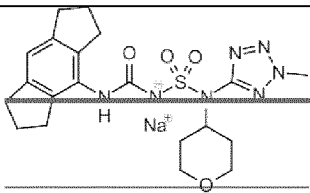
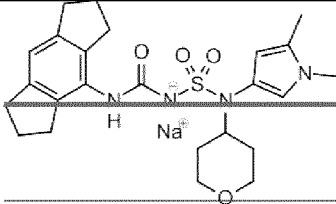
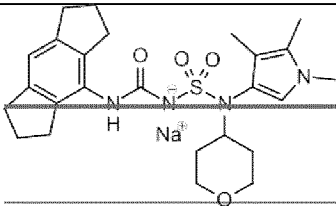
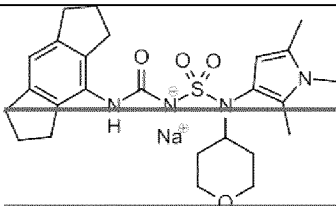
Compound No.	Structure
112	
113	
114	
115	
116	
117	

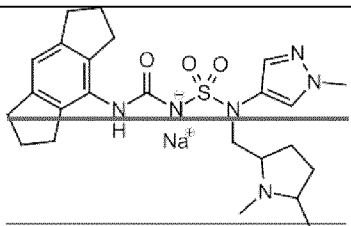
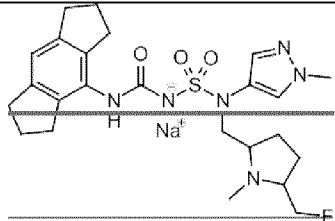
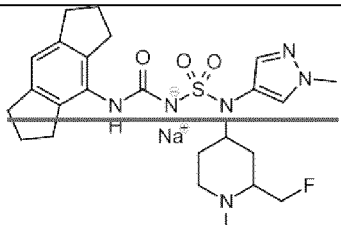
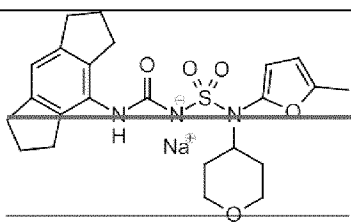
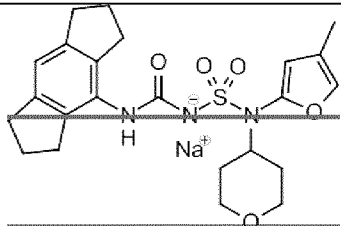
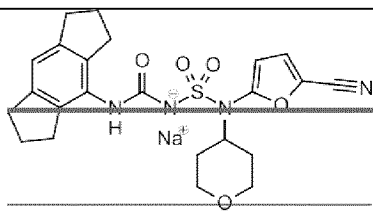
Compound No.	Structure
118	
119	
120	
121	
122	
123	

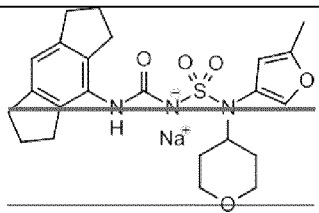
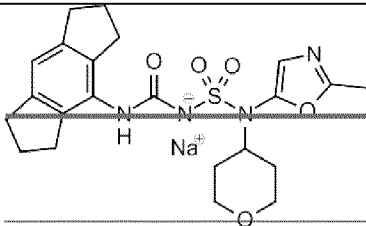
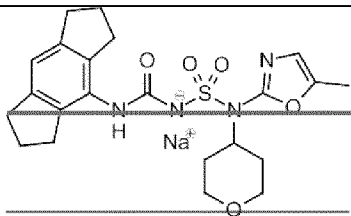
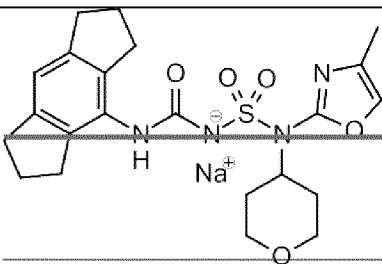
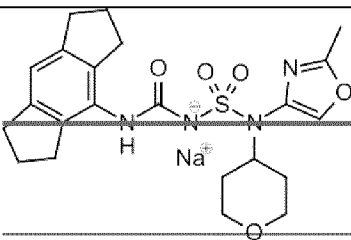
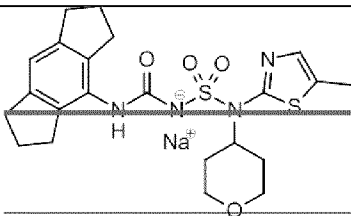
Compound No.	Structure
124	
125	
126	
127	
128	
129	
130	

Compound No.	Structure
131	
132	
133	
134	
135	
136	

Compound No.	Structure
137	
138	
139	
140	
141	
142	

Compound No.	Structure
143	
144	
145	
146	
147	
148	
149	

Compound No.	Structure
150	
151	
152	
153	
154	
155	

Compound No.	Structure
156	
157	
158	
159	
160	
161	

Compound No.	Structure
162	
163	
164	
165	
166	

and pharmaceutically acceptable salts thereof.

~~107~~. A pharmaceutical composition comprising the compound of any one of claims 1 to ~~96~~ or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable diluent or carrier.

~~118~~. The compound of any one of claims 1 to ~~96~~, or the pharmaceutical composition of claim ~~107~~,

for use in inhibiting inflammasome activity; optionally, the inflammasome is NLRP3 inflammasome, and the activity is *in vitro* or *in vivo*.

~~129~~. The compound of any one of claims 1 to ~~96~~, or the pharmaceutical composition of claim ~~107~~, for use in treating or preventing a disease or disorder; preferably wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder, an autoimmune disorder, a neurodegenerative disease, or cancer.

~~1310~~. The compound or pharmaceutical composition for use of claim ~~118~~ or claim 129, wherein the disease or disorder is an inflammatory disorder, an autoinflammatory disorder or an autoimmune disorder; optionally, the disease or disorder is selected from cryopyrin-associated autoinflammatory syndrome (CAPS; e.g., familial cold autoinflammatory syndrome (FCAS), Muckle-Wells syndrome (MWS), chronic infantile neurological cutaneous and articular (CINCA) syndrome/ neonatal-onset multisystem inflammatory disease (NOMID)), familial Mediterranean fever (FMF), nonalcoholic fatty liver disease (NAFLD), non-alcoholic steatohepatitis (NASH), gout, rheumatoid arthritis, osteoarthritis, Crohn's disease, chronic obstructive pulmonary disease (COPD), chronic kidney disease (CKD), fibrosis, obesity, type 2 diabetes, multiple sclerosis, dermatological diseases (e.g., acne) and neuroinflammation occurring in protein misfolding diseases (e.g., Prion diseases).

~~1411~~. The compound or pharmaceutical composition for use of claims ~~118~~ or claim 129, wherein disease or disorder is a neurodegenerative disease; optionally, the disease or disorder is Parkinson's disease or Alzheimer's disease.

~~1512~~. The compound or pharmaceutical composition for use of claims ~~118~~ or claim 129, wherein the disease or disorder is cancer; optionally, the cancer is metastasising cancer, gastrointestinal cancer, brain cancer, skin cancer, non-small-cell lung carcinoma, head and neck squamous cell carcinoma or colorectal adenocarcinoma.